Example 1.3: The standard error of the sample median of a normal distribution

For a normally distributed population with mean μ and variance σ^2, show that the standard error of the sample median, $\widehat{m}$, is given by

$$\text{s.e. } (\widehat{m}) = \sigma\sqrt{\frac{\pi}{2n}}.$$

Proof:

Given that the density function of a normal distribution with mean μ and variance σ^2 is

$$f(m) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{(m-\mu)^2}{2\sigma^2}}.$$

And from the symmetry of a normal distribution, we know that median is $m = \mu$.

So, $f(m) = \frac{1}{\sqrt{2\pi}\sigma}$.

Thus, $\text{s.e. } (\widehat{m}) = \frac{1}{2f(m)\sqrt{\mathrm{n}}}$

$$= \frac{1}{2\left(\frac{1}{\sqrt{2\pi}\sigma}\right)\sqrt{\mathrm{n}}}$$

$$= \sigma\sqrt{\frac{\pi}{2n}} \quad \text{(shown)}$$